THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

December 14, 2015 TIME: 9:30a.m.-12:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL TEN questions. Each question carries 10 marks.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- If you do not otherwise specify, you are assumed to be answering the questions in order. In particular, indicate on each question where your answer to the next question is if it does not follow immediately. (For example you may write 'turn back 3 pages for Question 5' at the end of Question 4.)
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.

1. Find all critical points of the function $w = x^3 + y^3 - 12x - 3y + 20$. Also determine the nature of the critical points (relative maximum, relative minimum or saddle point).

2. Find the shortest distance from the the origin to intersection of $z^2 = x^2 + y^2$ and $x - 2z = 3$.

3. Using ONE iteration of Newton- Raphson's method, find the first approximate for the solution to the system of equations

$$\begin{cases} x^2 + y^2 = 4 \\ y + e^x = 1 \end{cases}$$

with initial guess $(1, -1.7)$. (Round your answers to 4 decimal places.)

4. Evaluate the double integral $\int\int_{\Omega} (x+y)dxdy$, where Ω is the region between $y = x^3$ and $y = x^4$, $x \in [-1, 1]$.

5. Evaluate the double integral $\int\int_{D} dxdy$, where D is the region in the first quadrant bounded by $xy = 4$, $xy = 8$, $xy^3 = 5$ and $xy^3 = 15$.

6. Find the centroid ($\delta = 1$) of the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$, and bounded below by the $xy-$plane.

7. Use spherical coordinate to evaluate

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{0}^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} dxdydz.$$

8. Solve the system of linear equations

$$\begin{cases} x_1 + 2x_2 + 3x_3 = -1 \\ 3x_1 + 8x_2 + 11x_3 = 1 \\ x_1 + 5x_2 + 7x_3 = 1 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

9. Consider $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 5 \\ 5 & 8 & -5 & 12 \\ 3 & 16 & -11 & 28 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Determine conditions on b_1, b_2 and b_3 such that the system of linear equations is consistent.

(b) Solve the linear system when $\mathbf{b} = \begin{bmatrix} 1 & 4 & 4 \end{bmatrix}^T$.

10. Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a nonsingular matrix P such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate A^{2015}.

******* **END OF PAPER** *******